adapted on
Lynford L. Goddard
Brian T. Cunningham

Lectures 4

Review of Vector Calculus

Divergence and Curl

Maxwell's Equations in Differential Form

Victoria Shao
yangshao@Illinois.edu

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Fundamental Theorem of Single Variable Calculus

$$\int_a^b f'(x)dx = f(b) - f(a)$$

df=f’(x)dx is the infinitesimal change of f
in going from x to x+dx

Thus, chopping up the interval (a,b) into pieces dx
and adding up df gives the total change

Fundamental Theorem of Multi-Variable Calculus

$$\int_a^b \nabla f \bullet d\vec{l} = f(b) - f(a)$$

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \qquad \nabla f = \frac{\partial f}{\partial x}\hat{a}_x + \frac{\partial f}{\partial y}\hat{a}_y + \frac{\partial f}{\partial z}\hat{a}_z$$

df= ∇f·d**l** is the infinitesimal change of f
in going from (x,y,z) to (x+dx,y+dy,z+dz)

Thus, chopping up the path (a,b) into pieces d**l**
and adding up df gives the total change

∇f is a conservative field

$$\int_a^b \nabla f \bullet d\vec{l} = f(b) - f(a)$$

$$\nabla f = \frac{\partial f}{\partial x}\hat{a}_x + \frac{\partial f}{\partial y}\hat{a}_y + \frac{\partial f}{\partial z}\hat{a}_z$$

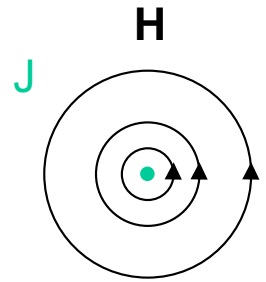
The right hand side doesn't depend on path so

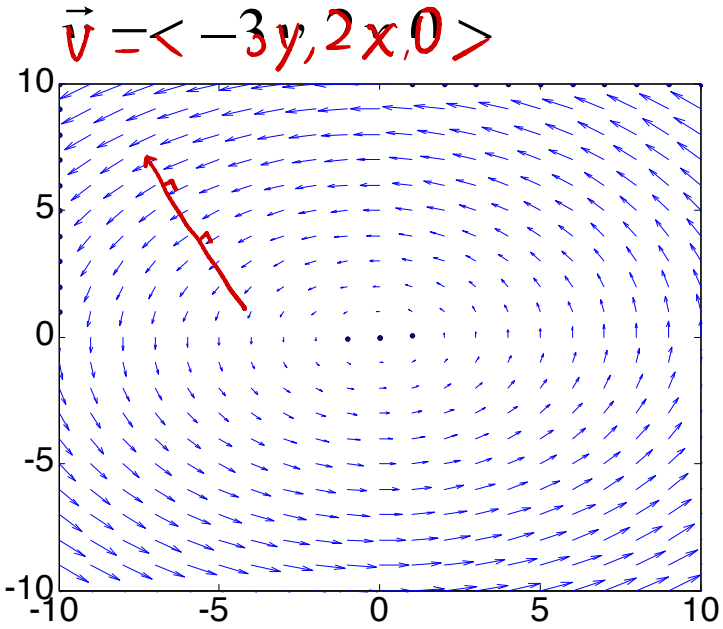
∇f is conservative

$$\oint_C \nabla f \bullet d\vec{l} = 0$$

∇f is curl-free

Curl





$$\nabla \times \vec{v} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = 5\hat{a}_z$$

$$= \hat{a}_x \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right)$$

$$-\hat{a}_y \left(\frac{\partial v_z}{\partial x} - \frac{\partial v_x}{\partial z} \right)$$

$$+\hat{a}_z \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right)$$

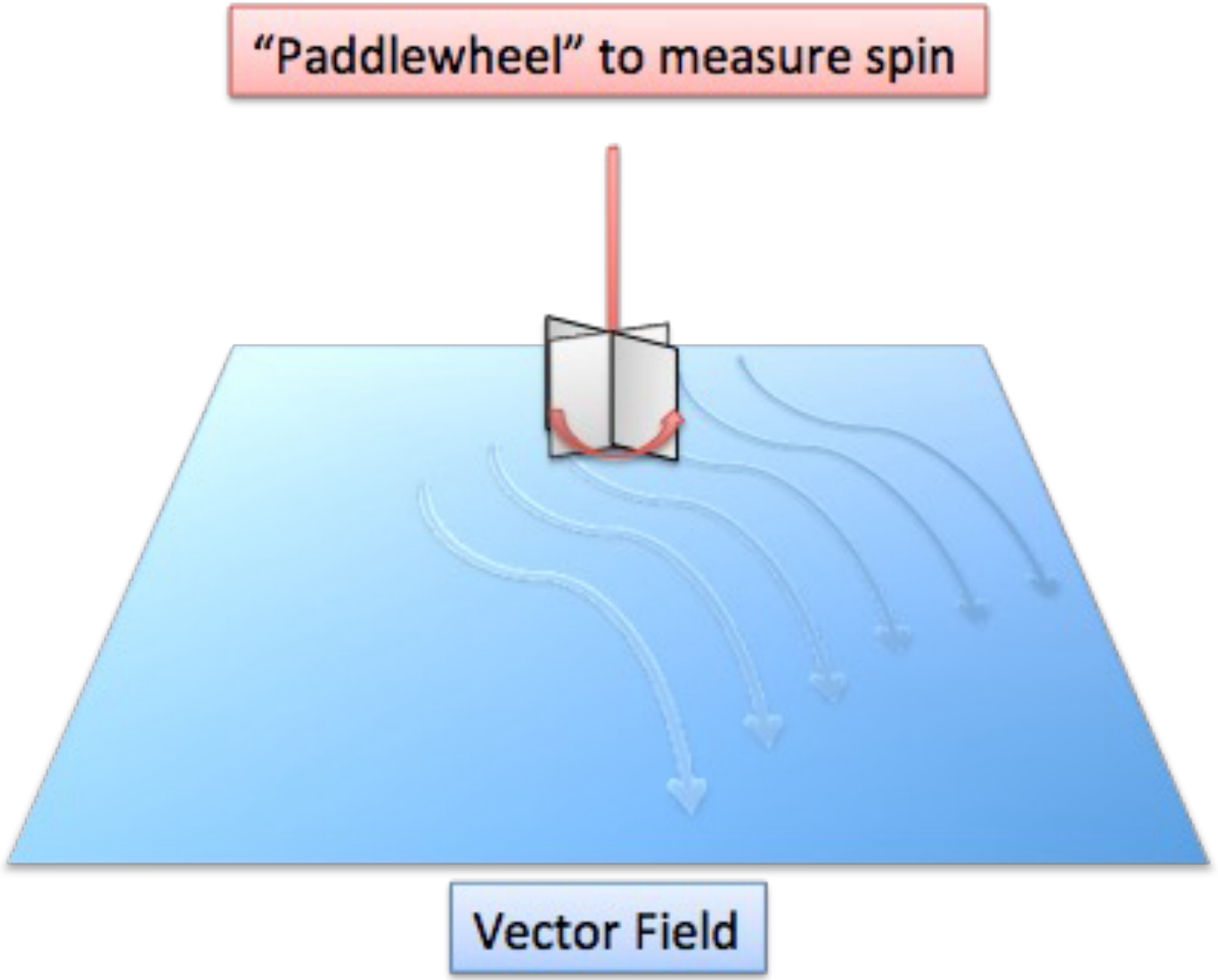
circulation per unit area

$\nabla \times \vec{v}$ is a measure of how much the vector field $\vec{v}$ circulates at a given point

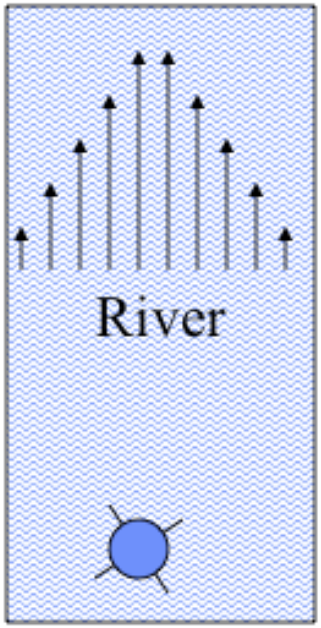
A place of high curl is like a whirlpool

– Everywhere here is a whirlpool of strength 5

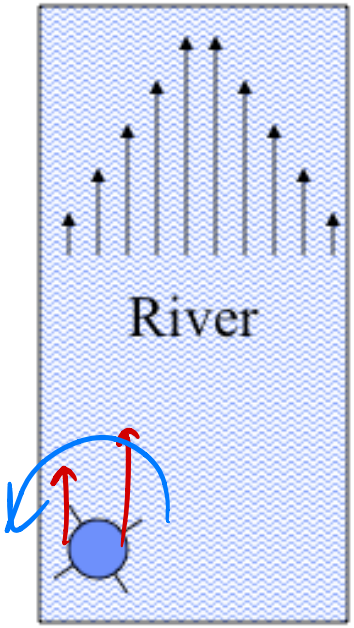
Physical Interpretation of Curl



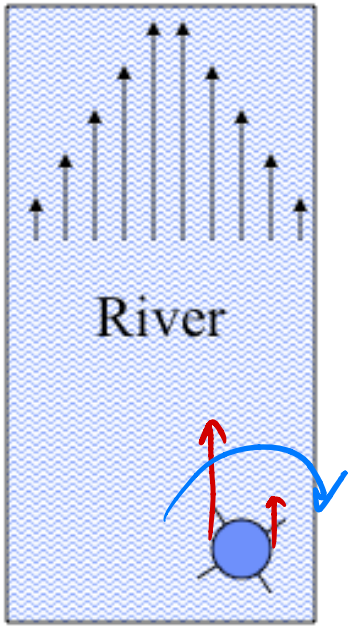
Physical Interpretation of Curl



No rotation!

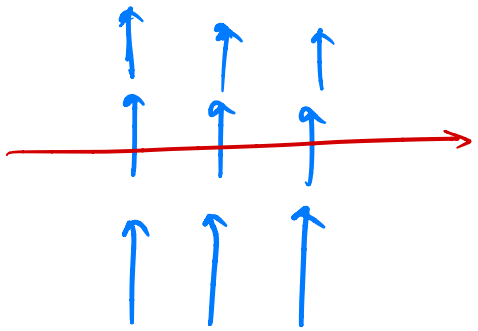


Anti-clockwise rotation.



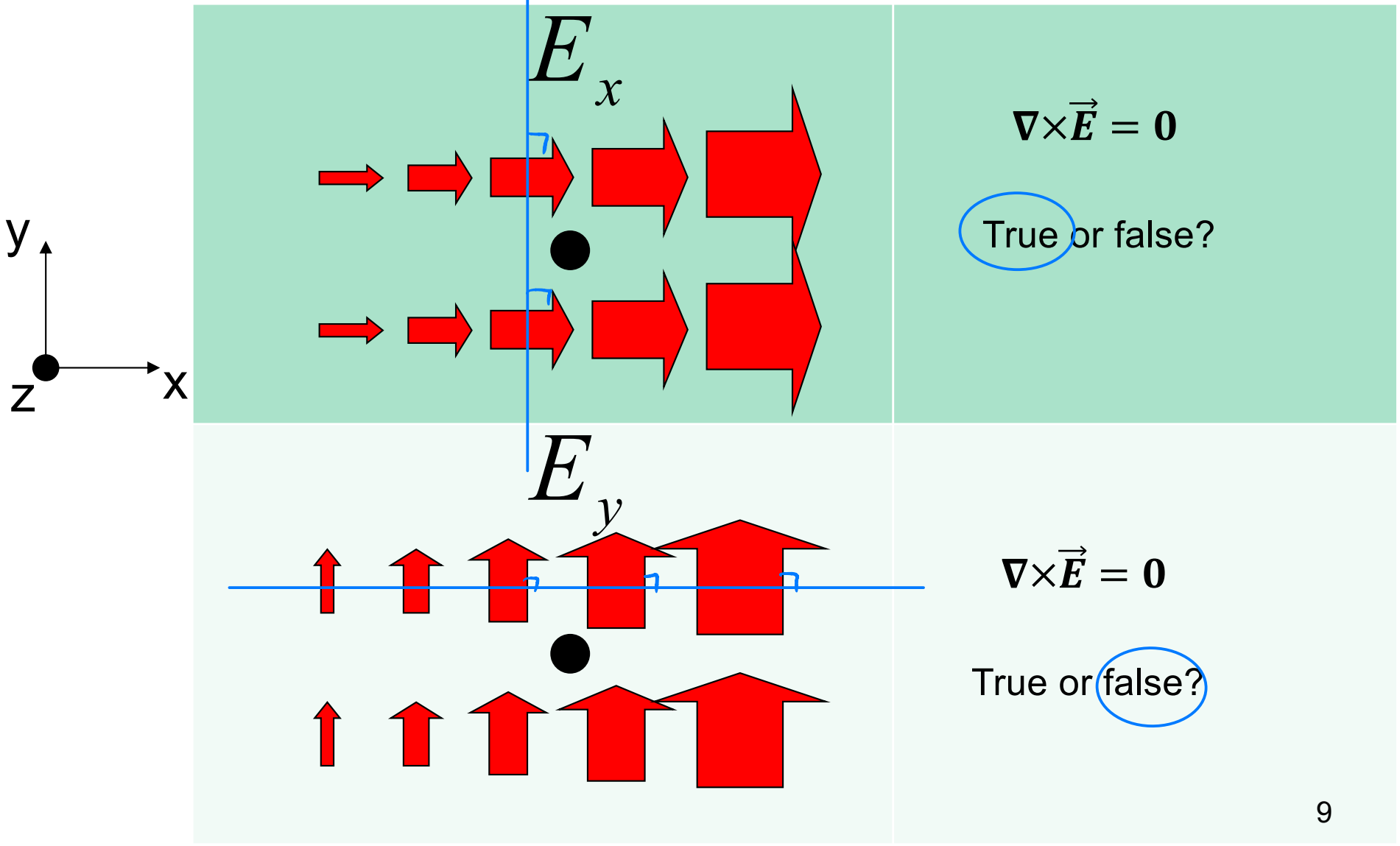
Clockwise rotation.

Meaning of Curl



- The curl meter only spins if there is a non-uniformity in the vector field in a direction **perpendicular** to the field
  - Curl describes variation ACROSS the flow of the field
- Rotation rate is proportional to the degree of non-uniformity
- Rotation is described with a magnitude and a direction – so it's a VECTOR and it's given by the right hand rule

Curl



$$\nabla \times \vec{E} = 0$$

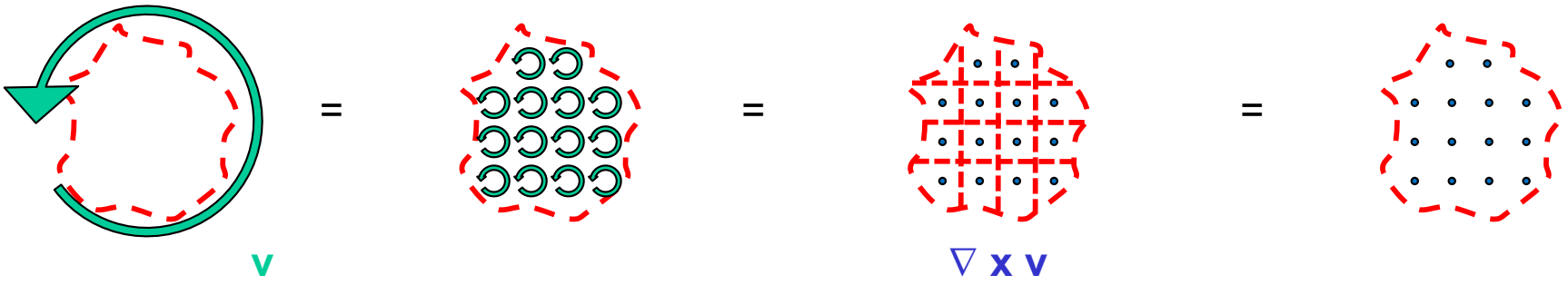
True or false?

$$\nabla \times \vec{E} = 0$$

True or false?

Stokes' Theorem

$$\oint_C \vec{v} \bullet d\vec{l} = \sum \oint_{C_i} \vec{v} \bullet d\vec{l} = \sum \iint_{S_i} (\nabla \times \vec{v}) \bullet d\vec{S} = \iint_S (\nabla \times \vec{v}) \bullet d\vec{S}$$



∇ x **v** is a measure of how much the vector field **v** circulates at a given point

A place of high curl is like a whirlpool

Thus, adding up the circulation from each whirlpool inside a region = the total circ. along the boundary

v is a conservative field iff ∇ x **v** = 0

$$\oint_C \vec{v} \bullet d\vec{l} = 0 = \iint_S (\nabla \times \vec{v}) \bullet d\vec{S}$$

The following are therefore equivalent:

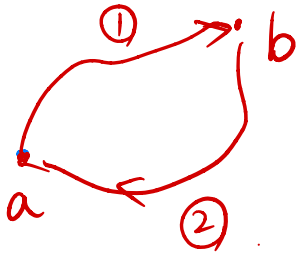
v is conservative

$$\oint_C \vec{v} \bullet d\vec{l} = 0$$

$$\vec{v} = -\nabla f \ \text{for some scalar potential f}$$

v is curl-free

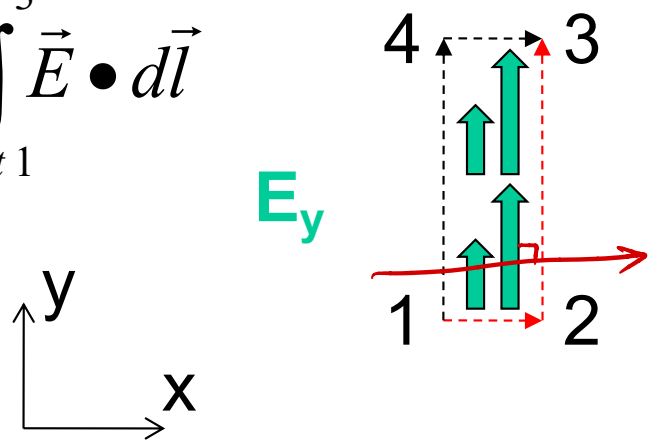
$$\int_a^b \vec{v} \bullet d\vec{l} \ \text{is path independent}$$



Curl at a Point

"E has CURL" means $EMF \neq 0$ around a tiny closed path at a particular point – line integral is path dependent

$$\int_{Pt\,1}^{Pt\,3} \vec{E} \bullet d\vec{l}$$



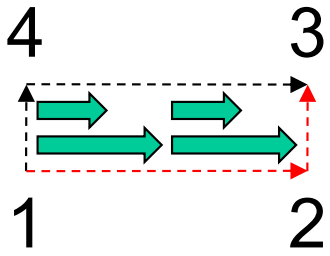
E_y

If there is any DIFFERENCE in E_y in the x direction

$$\frac{dE_y}{dx} \neq 0$$

Path 123 ≠ Path 143

E_x



Or if there is any DIFFERENCE in E_x in the y direction

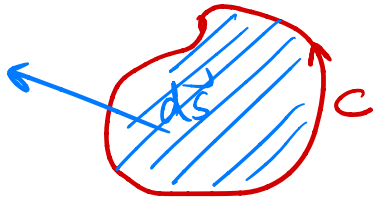
$$\frac{dE_x}{dy} \neq 0$$

Path 123 ≠ Path 143

Maxwell's Equations in Differential Form

Faraday's Law

$$EMF = \oint_C \vec{E} \bullet d\vec{l} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S}$$



Stokes' Thm

$$\oint_C \vec{E} \bullet d\vec{l} = \iint_S (\nabla \times \vec{E}) \bullet d\vec{S} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S}$$

$$\Rightarrow \boxed{\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}}$$

Maxwell's Equations in Differential Form

Ampere's Law

$$\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt} \iint_S \vec{D} \bullet d\vec{S}$$

Static

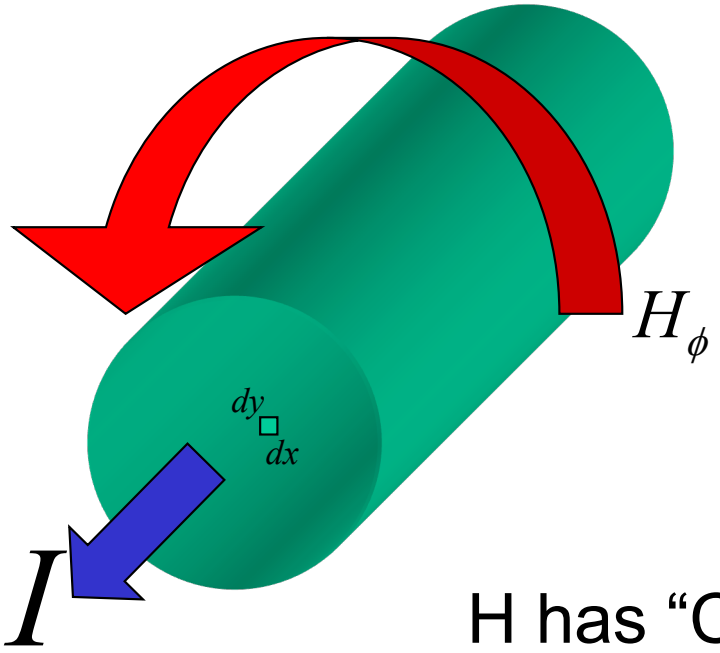
dynamic

Stokes' Thm

$$\oint_C \vec{H} \bullet d\vec{l} = \iint_S (\nabla \times \vec{H}) \bullet d\vec{S} = \iint_S (\vec{J} + \frac{d\vec{D}}{dt}) \bullet d\vec{S}$$

$$\Rightarrow \boxed{\nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}}$$

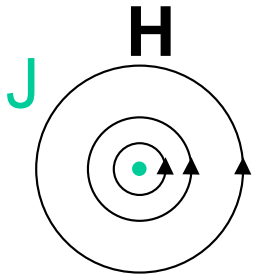
Ampere's Law In Differential Form



$$MMF = \oint \vec{H} \bullet d\vec{l} \neq 0$$

If there is any current going through a particular point

H has "CURL" at a point if there is current going through the point:

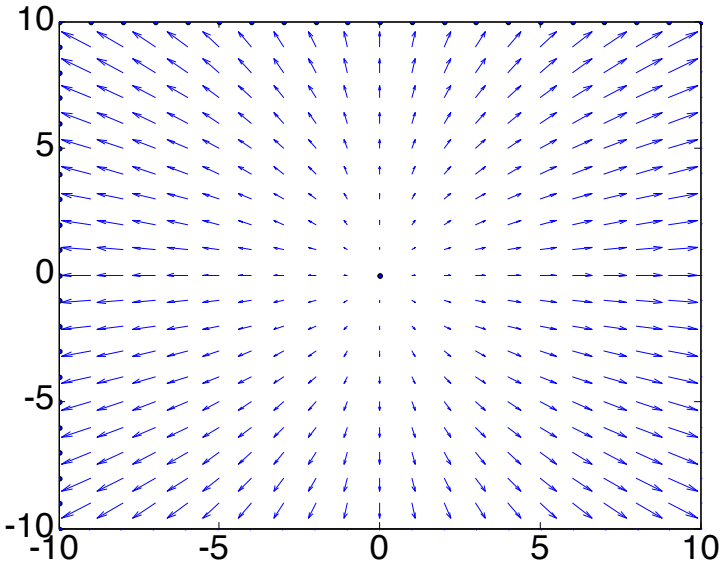


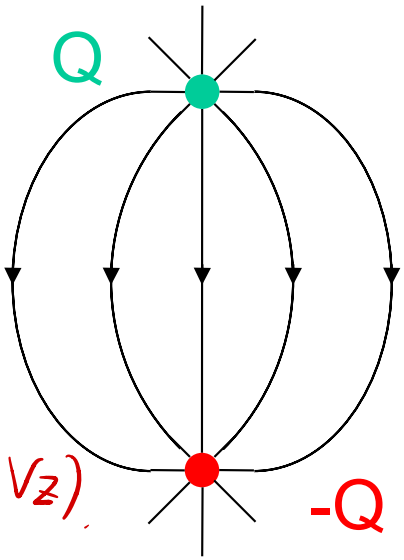
CONDUCTION current → J

DISPLACEMENT current → $\frac{d\vec{D}}{dt}$

Divergence

$$\vec{v} = \langle 3x, 2y, 0 \rangle$$





$$\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \cdot (V_x, V_y, V_z)$$

$$\nabla \bullet \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 5$$

flux per unit volume

$\nabla \cdot \vec{v}$ is a measure of how many field lines for **v** are created at a given point

A place of high divergence is like a water faucet

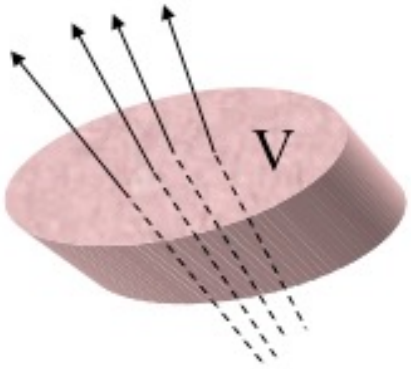
— Everywhere here is a faucet of strength 5



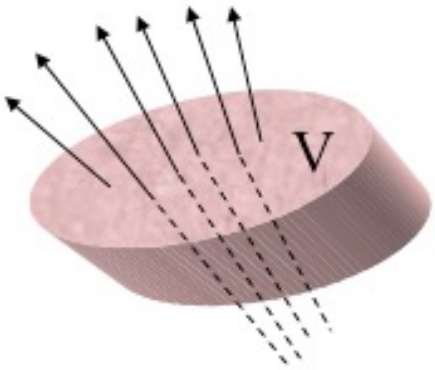
$$\text{Divergence} = \lim_{\text{Vol}\to 0} \frac{\text{Flux}}{\text{Vol}}$$

Meaning of Divergence

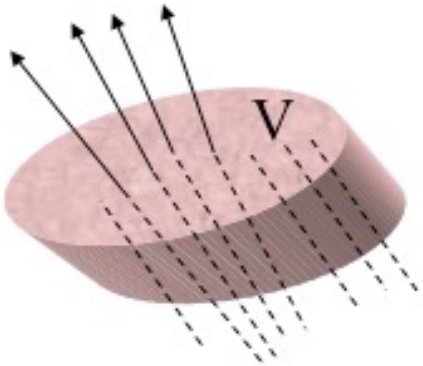
$$\text{Divergence} = \lim_{V \to 0} \frac{\text{Flux}}{V}$$



Flux in = flux out so no sources or sinks inside V.

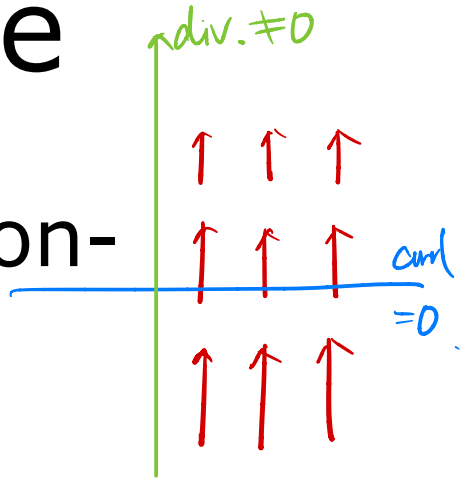


Flux out > flux in Positive divergence. Must be a source inside V.



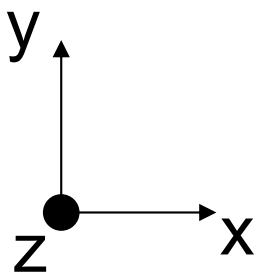
Flux out < flux in Negative divergence. Must be a sink or drain inside V.

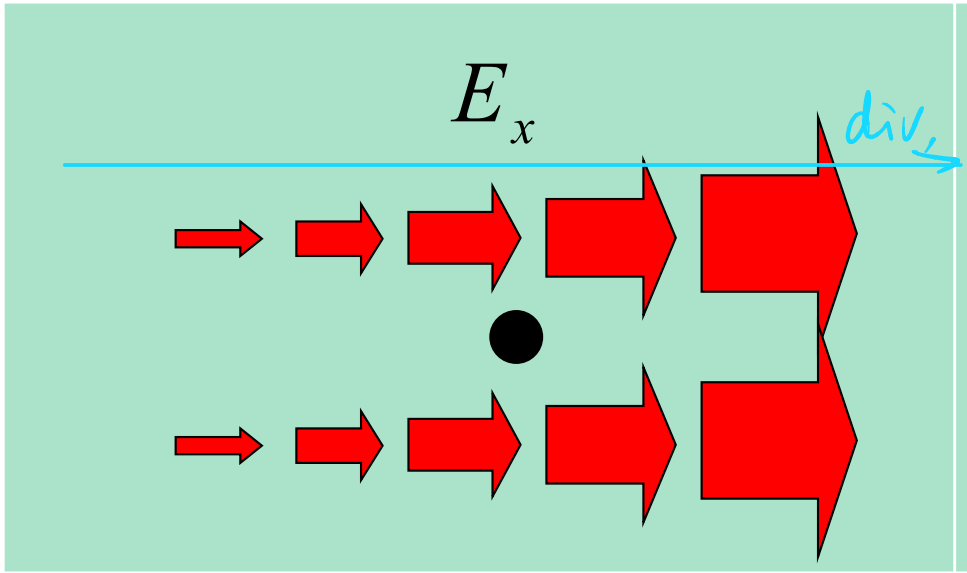
Meaning of Divergence



- There is divergence if there is a non-uniformity in the vector field in a direction **parallel** to the field
  - Divergence describes variation ALONG the flow of the field
- Divergence is proportional to the degree of non-uniformity
- Divergence is described only by the magnitude – so it's a SCALAR

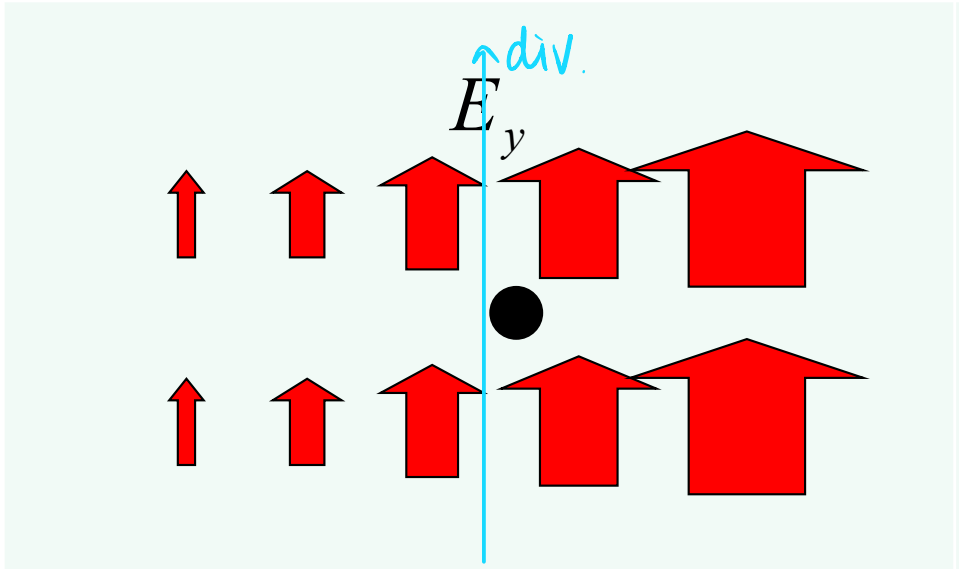
Divergence





$$\nabla \cdot \vec{E} = 0$$

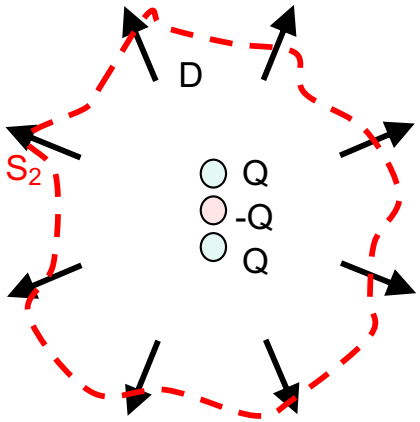
True or false?



$$\nabla \cdot \vec{E} = 0$$

True or false?

Divergence Theorem



$$\int_V \nabla \cdot \vec{\boldsymbol{D}}\, dV = \oint_S \vec{\boldsymbol{D}} \cdot d\vec{\boldsymbol{S}}$$

For a general vector field $\vec{\boldsymbol{v}}$

$$\iiint_V \nabla \bullet \vec{v}\; dV = \oiint_{\partial V} \vec{v} \bullet d\vec{S}$$

$\nabla \cdot \vec{\boldsymbol{v}}$ is a measure of how many field lines for $\vec{\boldsymbol{v}}$ are created at a given point

Thus, adding up the net # lines created inside a volume = the flux of lines out its boundary

Maxwell's Equations in Differential Form

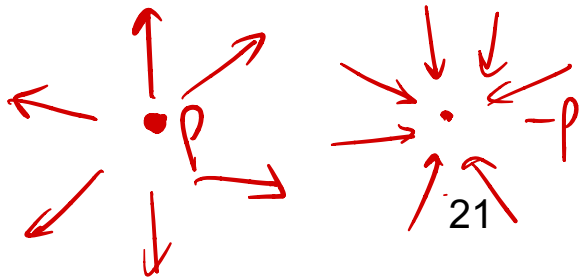
Gauss' Law

$$\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \rho dV = Q$$

Divergence Thm

$$\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \nabla \bullet \vec{D}\ dV = \iiint_V \rho dV$$

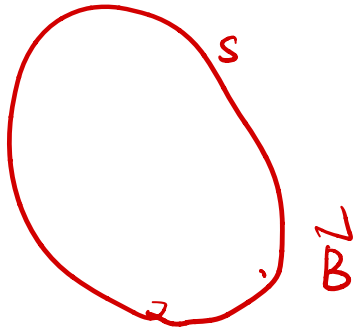
$$\Rightarrow \nabla \bullet \vec{D} = \rho$$



Maxwell's Equations in Differential Form

Gauss' Law

$$\oiint_S \vec{B} \bullet d\vec{S} = 0$$



Divergence Thm

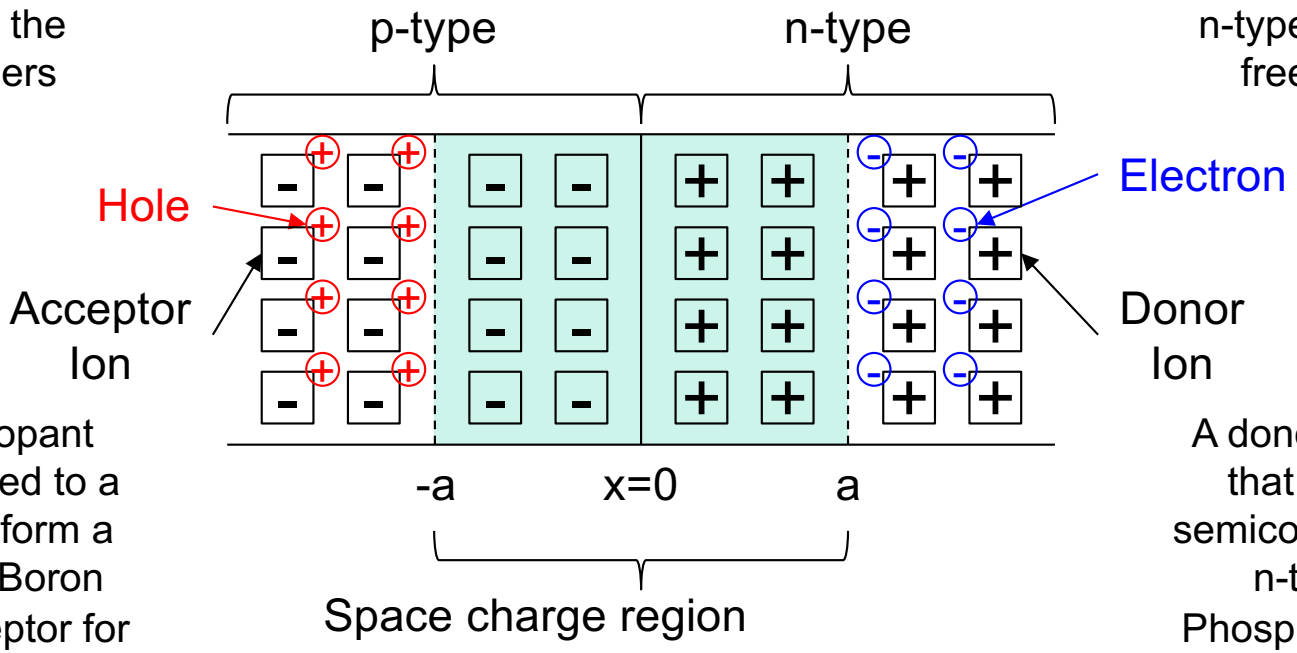
$$\oiint_S \vec{B} \bullet d\vec{S} = \iiint_V \nabla \bullet \vec{B}\ dV = 0$$

$$\Rightarrow \boxed{\nabla \bullet \vec{B} = 0}$$

Differential form of Gauss's law for magnetism

E-field of pn junction (ECE 340)

p-type: holes are the free charge carriers



n-type: electrons are the free charge carriers

An acceptor is a dopant atom that when added to a semiconductor can form a p-type region, e.g. Boron (group III) is an acceptor for Silicon (group IV)

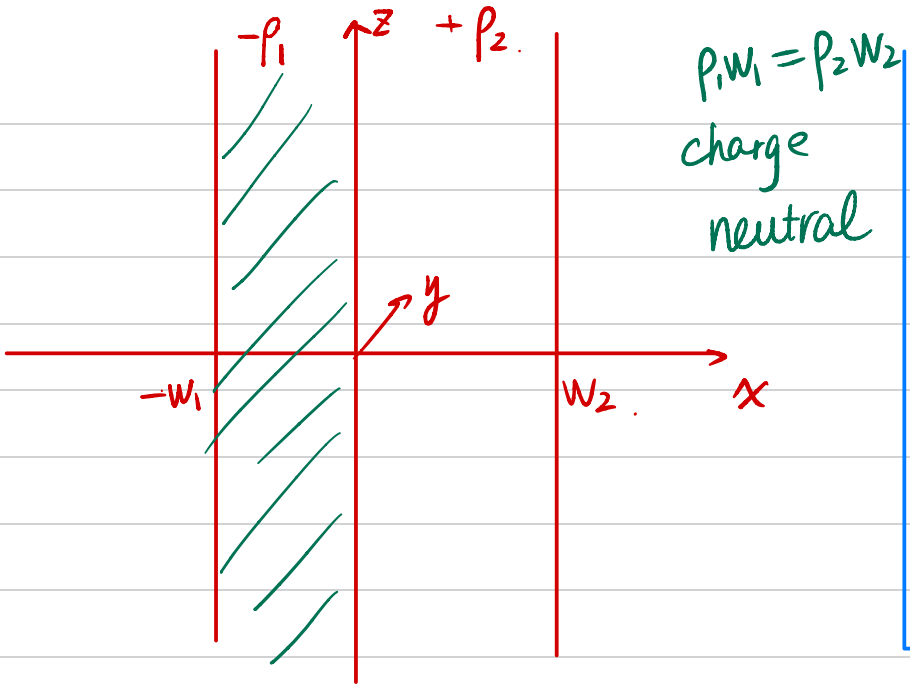
A donor is a dopant atom that when added to a semiconductor can form an n-type region, e.g. Phosphorus (group V) is a donor for Silicon (group IV)

- Find the **E**-field for an evenly doped: $N_d=N_a$ pn junction: $\rho=\{ -\rho_0=-qN_a$ for $-a<x<0$,

$$\{ \ \rho_0=qN_d \ \text{for } 0<x<a,$$

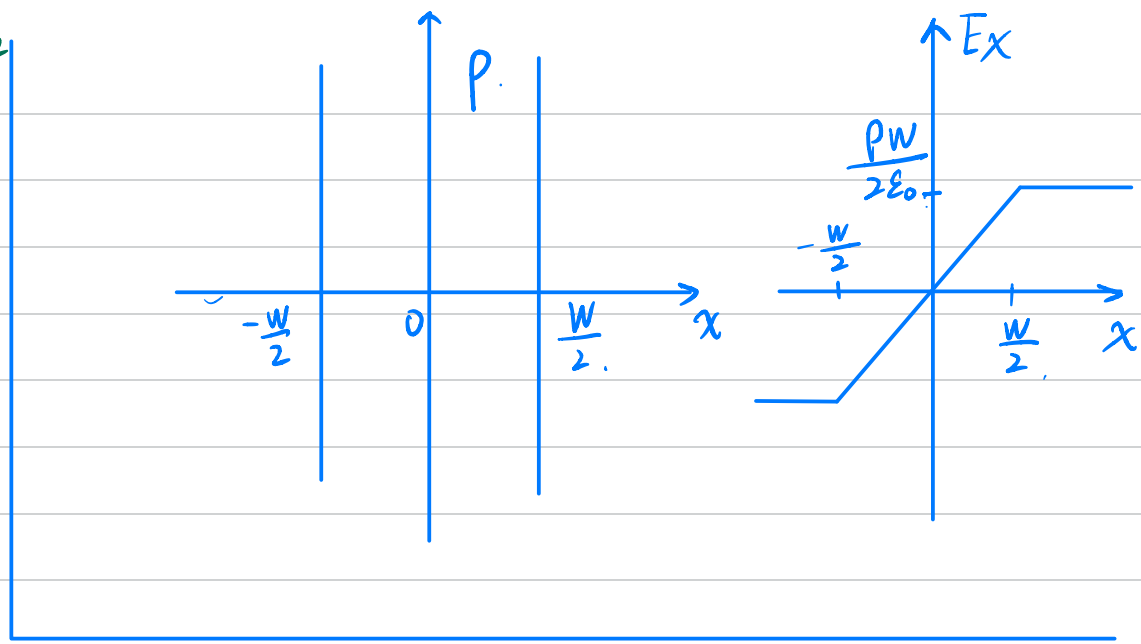
$$\{ \quad 0 \quad \text{for } |x|>a$$

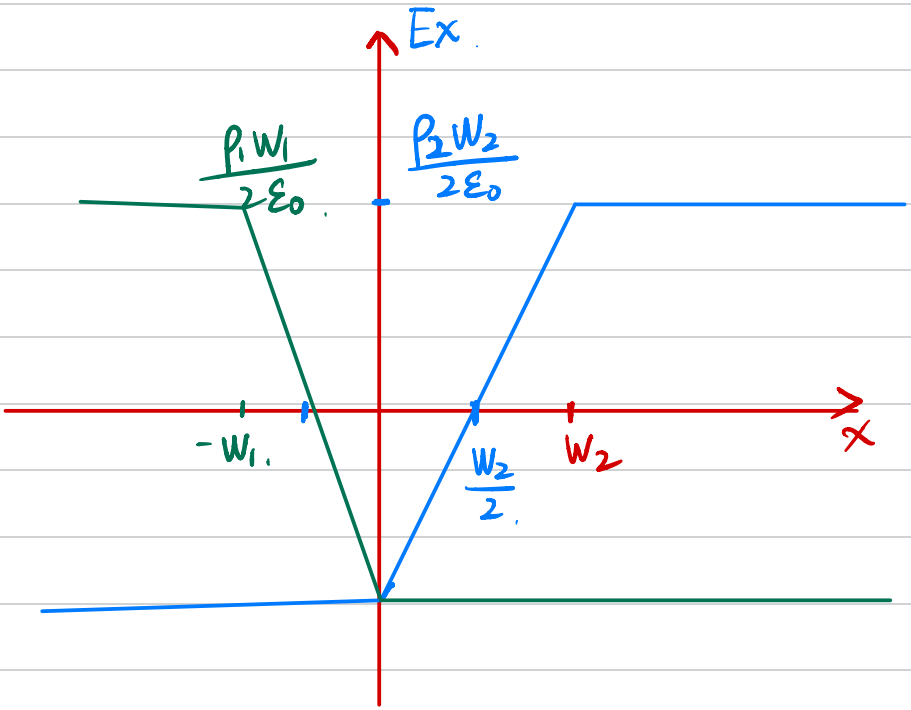
From Example 3.5 (p 145) of old book

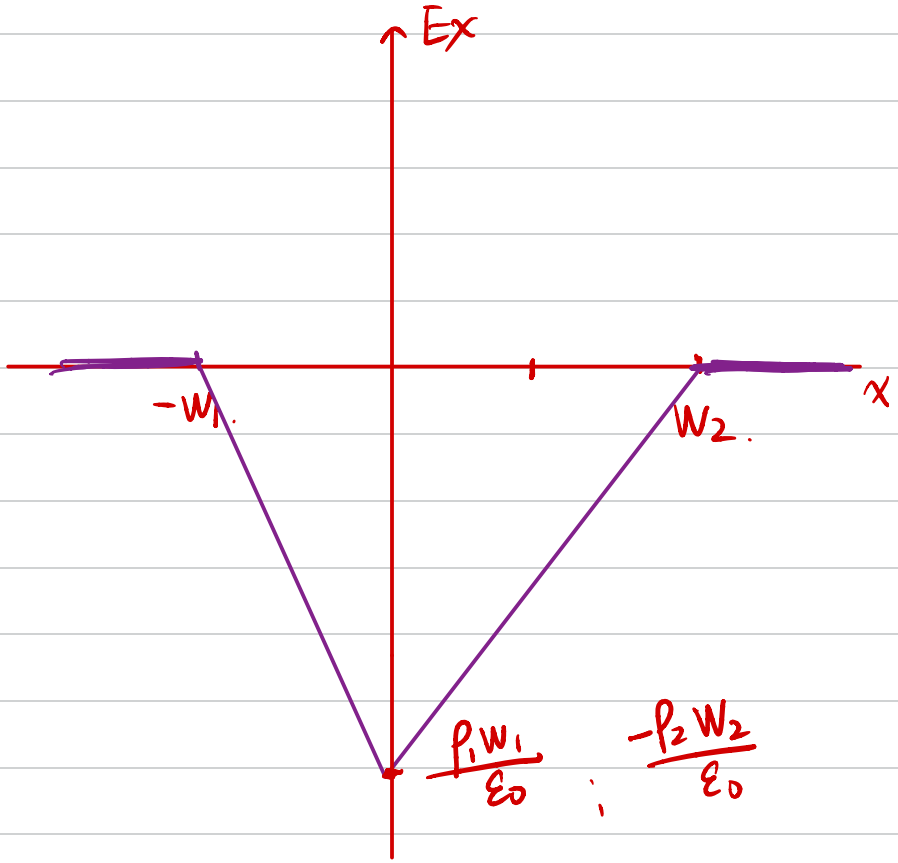


$\rho_1 w_1 = \rho_2 w_2$

charge neutral



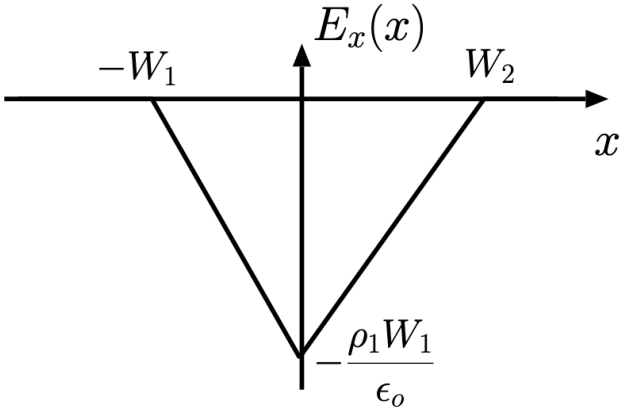




Example: Divergence

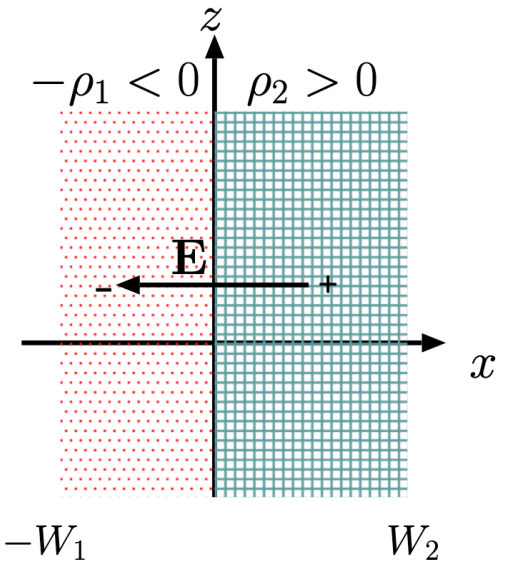
Find the divergence $\nabla \cdot \vec{E}$ of electric field vector

$$\mathbf{E} = \begin{cases} -\hat{x}\frac{\rho_1(x+W_1)}{\epsilon_o}, & \text{for } -W_1 < x < 0 \\ \hat{x}\frac{\rho_2(x-W_2)}{\epsilon_o}, & \text{for } 0 < x < W_2 \\ 0, & \text{otherwise,} \end{cases}$$



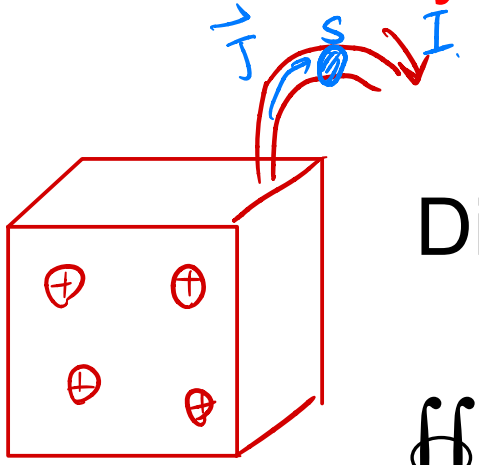
$$\nabla \cdot \vec{E} = \left(\frac{\partial}{\partial x}\hat{x}, \frac{\partial}{\partial y}\hat{y}, \frac{\partial}{\partial z}\hat{z}\right) \cdot (E_x, E_y, E_z)$$

$$= \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}.$$



Continuity Equation in Differential Form

Continuity Eqn



$$I = \oiint_S \vec{J} \bullet d\vec{S} = -\frac{d}{dt}\iiint_V \rho dV$$

Divergence Thm

$$\oiint_S \vec{J} \bullet d\vec{S} = \iiint_V \nabla \bullet \vec{J}\ dV = -\frac{d}{dt}\iiint_V \rho dV$$

$$\Rightarrow \nabla \bullet \vec{J} = -\frac{d\rho}{dt}$$



Useful Relationships

$$\nabla \bullet (\nabla \times \vec{A}) = 0$$

$$\nabla \times (\nabla f) = 0 \quad \text{We already knew } \nabla f \text{ is conservative}$$

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \bullet \vec{A}) - \nabla^2 \vec{A} \quad \text{Must use cartesian coordinates!}$$

$$\nabla^2 \vec{A} \equiv (\nabla^2 A_x)\hat{x} + (\nabla^2 A_y)\hat{y} + (\nabla^2 A_z)\hat{z}$$

$$0 = \nabla \bullet (\nabla \times \vec{H}) = \nabla \bullet (\vec{J} + \frac{d\vec{D}}{dt})$$

$$= \nabla \bullet \vec{J} + \frac{d}{dt}(\nabla \bullet \vec{D}) = \nabla \bullet \vec{J} + \frac{d\rho}{dt}$$

$$\therefore \nabla \bullet \vec{J} = -\frac{d\rho}{dt}$$

The continuty eqn. is contained in Maxwell's Eqns.

Summary of Maxwell's Equations

Faraday's Law $\oint_C \vec{E} \bullet d\vec{l} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S}$ $\quad \nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$

Ampere's Law $\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt}\iint_S \vec{D} \bullet d\vec{S}$ $\quad \nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$

Gauss' Law $\oiint_S \vec{B} \bullet d\vec{S} = 0$ $\quad \nabla \bullet \vec{B} = 0$

Gauss' Law $\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \rho dV$ $\quad \nabla \bullet \vec{D} = \rho$

Continuity Eq. $\oiint_S \vec{J} \bullet d\vec{S} = -\frac{d}{dt}\iiint_V \rho dV$ $\quad \nabla \bullet \vec{J} = -\frac{d\rho}{dt}$

Laws of Electrostatics

- All static electric fields E, obtained from Coulomb's law, and satisfying Gauss's law $\nabla \cdot \vec{\boldsymbol{D}} = \rho$ with static charge densities $\rho = \rho(r)$, are also found to be curl-free without exception.

Electrostatics
$\nabla \cdot \vec{\boldsymbol{D}} = \rho$
$\nabla \times \vec{\boldsymbol{E}} = 0$
$\vec{\boldsymbol{D}} = \epsilon_0 \vec{\boldsymbol{E}}$

Helmholtz theorem